

Final Assignment - Advanced Time Series

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1 Objectives

The goal of this project is to replicate the main results of *Patent-based News Shocks* (Cascaldi-Garcia & Vukotić, 2022). This paper identifies aggregate and industry-level patent-based news shocks by exploiting changes in stock market valuations of firms that obtain patent grants and finds that the patent shocks resemble diffusion news: they do not affect total productivity in the short run but induce a strong permanent effect after five years. It also finds that patent-based news shocks produce positive co-movement between consumption, output, investment, and hours. Regarding methodology, it employs a Bayesian VAR framework to examine their effects on total factor productivity and the business cycle. It also includes a Bayesian proxy-SVAR as an alternative empirical strategy in the robustness analysis. This study is particularly interesting, as it introduces a novel perspective on technology shocks and incorporates several empirical methods that were discussed in class, providing multiple replication avenues for the assignment. For additional robustness checks, I also compare the results from standard local projections and Bayesian local projections to further validate the paper's findings.

The remainder of the paper is organized accordingly. Section 2 illustrates the resources and the process of the data. Section 3 reviews the methodology used in the literature, including Bayesian VAR, Bayesian Proxy SVAR, Local Projections and Bayesian Local Projections. Section 4 shows the main empirical result of this paper. Section 5 shows one robustness check of the paper, by changing the identification strategy to local projections, and identifying the patent news shock as instrument variable. Section 6 provides other useful discussions in the paper. Section 7 concludes.

2 Data Descriptions

2.1 Patent-Based News Shocks

The news literature usually defines technological news shocks as advance information about technology that will become available with some delay. Rarely, however, does this literature provide specific examples of such technological advances. Based on that, Cascaldi-Garcia and Vukotić (2022) identify technological news shocks using a measure, proposed by KPSS, that combines microlevel data on patents with their stock market valuation. It argues that unexpected innovations in this measure represent technological news shocks and refer to them as patent-based news shocks.

The KPSS aggregate innovation index is constructed by using rich firm-level data set, and estimates the patent's economic value by combining data on patents issued to US firms during the 1926–2010 period with firm stock price movements. In particular, by merging Google Patents with the Centre for Research in Security Prices (CRSP) database, they obtain a database of 1,928,123 patents and subsequent stock price movements. The biggest challenge the authors face is to carefully extract information about the patent's economic value contained in stock prices from unrelated news. To do so, they focus their analysis on the days around an announcement of a patent grant. These particular days are also characterized by larger trading activity in the stock of the firm. They then filter the stock price reaction to the patent issuance from noise by making several distributional assumptions; the results prove to be quite robust to these assumptions.

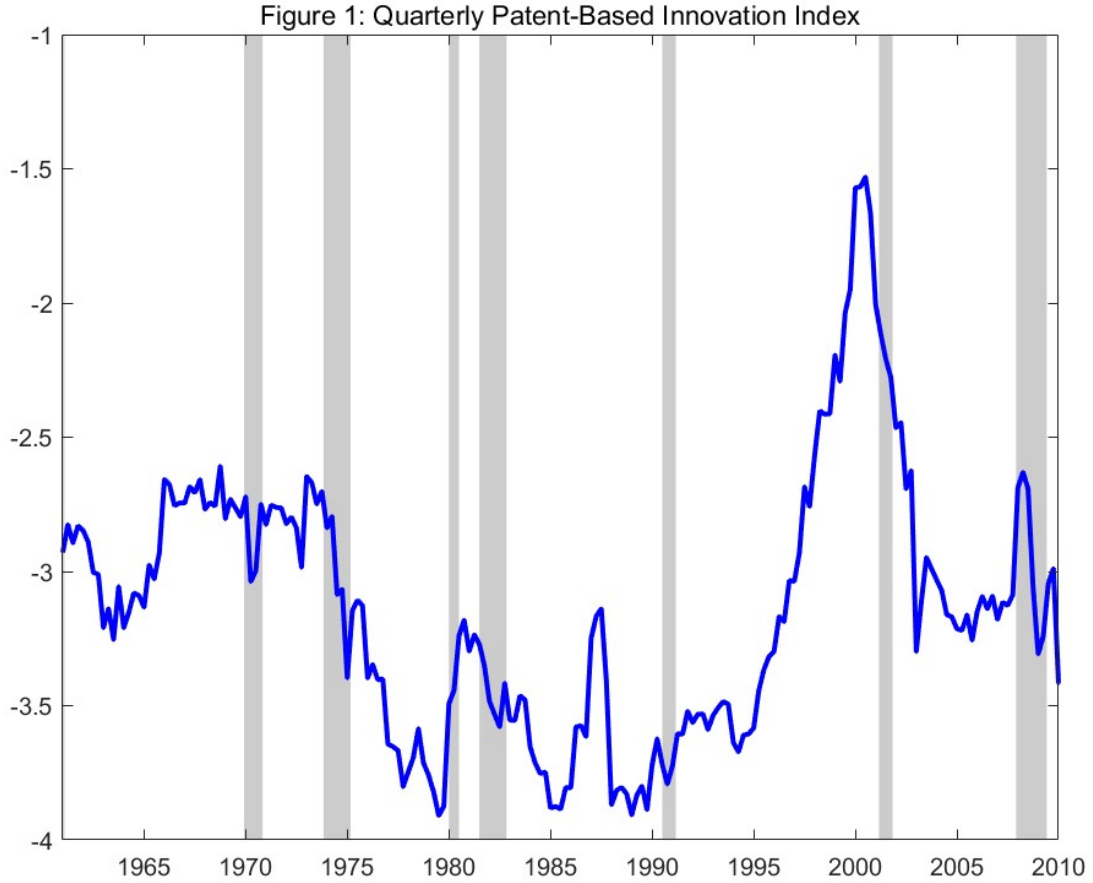


Figure 1: Aggregate patent-based quarterly index

Crucially for the analysis, the author construct a measure in a quarterly frequency, as displayed in Figure 1, and study the period after 1961:Q1 to account for data availability on consumer confidence, an essential variable in the news literature. They refer to this measure as the patent-based innovation index.

It is worth noting that a few reasons lead the authors to believe that the patent-based innovation index can be used to extract technological news shocks. First, by combining patent counts with stock market data, this measure overcomes the criticisms that arise when only simple patent counts are used for economic analysis. Second, the lag between a patent grant and when the product or process is brought to markets is worrisome if one is interested in recovering surprise technological shocks but it is a desirable feature if one is interested in recovering technological news shocks. Third, any plausible identification of technological news shocks must rely on the usage of forward-looking variables because of their predictive power regarding future movements in economic activity. Fourth, using microlevel data on patents and their stock market evaluations allows to propose an alternative, direct way to identify news shocks, independent of movements in TFP. Overall, rather than focusing on proposing a novel statistical procedure that would recover news shocks by exploiting fluctuations in TFP, the authors approach this problem differently by using microlevel data on patents and their stock market valuations.

The patent-based news shock is identified under a conventional lower triangular Cholesky decomposition, where the patent-based innovation index is ordered first in the information set. The use of a VAR is appropriate in this context because it purges the patent-based technology index from any lagged predictability. In fact, none of the endogenous variables predict the filtered patent-based technology index, i.e., patent-based news shock. In the literature, the authors argue that: First, none of the endogenous variables Granger cause patent-based news shocks. Second, the correlations between our patent-based news shock and other structural shocks in

the literature —news about tax shocks, oil price shocks, monetary policy shocks, and tax shocks —are rather small and insignificant. The correlations (and p-values) are 0.12 (0.37), -0.06 (0.46), 0.08 (0.39), -0.10 (0.37), respectively. The shock is mildly positively correlated with news about government spending shock (0.20), but it is not statistically significant at a 5 percent level. This mild correlation might reflect a positive response of government spending to the exogenous technological improvements.

2.2 Other Variables

The authors estimate the benchmark VAR model with 10 endogenous variables: patent-based innovation index, utilization-adjusted TFP, output, consumption, investment, hours, inflation, the federal funds rate, consumer confidence, and the stock price index. All variables except inflation (reported in annualized percent) are in log levels and quarterly, from 1961:Q1 to 2010:Q4, and the model contains four lags and an intercept term.

3 Methodology

There are three empirical strategy used in this literature: Bayesian VAR, Bayesian Proxy VAR, and Local Projections. The paper used the Bayesian VAR as the baseline model, and the other two models are for robustness checks. What’s more, for comparison, I add the Bayesian Local Projection in the robustness checks.

Bayesian VAR: In order to deal with the large number of coefficients by taking advantage of Minnesota priors (Bańbura et al., 2010). Coverage bands for the impulse response graphs are computed using 1,000 draws from the posterior distribution. Since this paper deals with data in quarterly frequency, which limits the size of the sample, and we are also interested in the medium- and long-run effects of the shock (up to 20 quarters ahead, and not only up to the p-included lags), the authors opted for the VAR approach at first.

Bayesian Proxy VAR: As a different identification strategy, the patent-based index should be directly linked to what is theoretically considered a technological news shock. The idea here is to employ the proxy VAR procedure introduced by Stock and Watson (2012) and Mertens and Ravn (2013) to identify the patent-based news shock by relying on the patent-based innovation index series as an instrument, which is a noisy measure of the structural shock. In summary, the procedure consists of regressing the instrument against the residuals of a reduced-form VAR and using this information to infer the contemporaneous impact of the patent-based news shock on the macroeconomic variables.

Two conditions must be satisfied for the applicability of such a series as an instrument: relevance and exogeneity. The first condition implies that the potential instrument is correlated with the true underlying structural shock. While the true structural shock is not directly observed, this condition cannot be tested. However, the extensive discussion presented in Section 2 makes it confident about the intrinsic relationship between the patent-based innovation index and the patent-based news shock. The second condition of exogeneity implies that the potential instrument is not correlated with any other structural shock. In order to illustrate this condition, The authors perform an exogeneity test by calculating the correlation between the patent-based index and several identified shocks from the literature —namely, news about tax shocks, news about government defense spending, oil price shocks, monetary policy shocks, and tax shocks. The results of the exogeneity tests shows that the patent-based innovation index is not correlated with oil price shocks and is mildly correlated to news about tax and tax shocks, but these are not statistically significant. Based on these results, they conclude that the patent-based index series passes the exogeneity test and can be used as an instrument in the proxy VAR setup.

The method used in this literature is based on Caldara and Herbst (2019), and the estimation of the reduced-form is using a standard Minnesota priors similar to the Bayesian VAR. It employ the high relevance prior for

the standard deviation of the measurement error, which determines the tightness of the relationship between the proxy and the SVAR model.

Local Projections: Method proposed by Jordà (2005).

For each horizon h ahead, it estimates a different set of regressions as

$$y_{t+h} = \alpha^h + B_1^h y_{t-1} + \dots + B_p^h y_{t-p} + u_{t+h}^h.$$

Where y_t is the same set of endogenous variables as in the baseline estimated for the impulse responses in the paper. The local projections are then calculated as

$$\begin{aligned}\hat{B}_1^0 &= \gamma_1, \\ \text{LP}(t, h, \gamma_1) &= \hat{B}_1^h \gamma_1,\end{aligned}$$

And γ_1 is identified as a Cholesky recursive formulation, where the patent-based innovation index is ordered first.

Bayesian Local Projections: Method proposed by Ferreira et al. (2025), who cast local projections into a Bayesian quasi-likelihood framework to manage the bias-variance trade-off in impulse response estimation.

As the same above, for each horizon $h = 1, \dots, H$, BLP treats LPs as a sequence of predictive regressions,

$$y_{t+h} = C^{(h)} + A_1^{(h)} y_t + A_2^{(h)} y_{t-1} + \dots + A_{\bar{p}}^{(h)} y_{t-\bar{p}+1} + u_{t+h}^{(h)},$$

which can be written compactly as

$$y_{t+h} = A_0^{(h)} x_t + u_{t+h}^{(h)}, \quad x_t \equiv (1, y_t', y_{t-1}', \dots, y_{t-\bar{p}+1}')'.$$

BLP adopts an auxiliary Normal likelihood at each horizon,

$$u_{t+h}^{(h)} \overset{\text{aux}}{\sim} \text{i.i.d. } \mathcal{N}(0, \Sigma_u^{(h)}),$$

which is deliberately misspecified (multi-step forecast errors are typically serially correlated), but delivers conjugate posteriors for the pseudo-true parameters via a Normal–Inverse Wishart prior:

$$\begin{aligned}\Sigma_u^{(h)} &\sim \mathcal{IW}(\Psi_0^{(h)}, \nu_0^{(h)}), \\ \text{vec}(A_0^{(h)}) \mid \Sigma_u^{(h)} &\sim \mathcal{N}(\text{vec}(A_{0, \text{prior}}^{(h)}), \Sigma_u^{(h)} \otimes \Omega_0^{(h)}).\end{aligned}$$

Under this setup, the posterior remains Normal–Inverse–Wishart:

$$\begin{aligned}\Sigma_u^{(h)} \mid Y &\sim \mathcal{IW}(\Psi^{(h)}, \nu^{(h)}), \\ \text{vec}(A_0^{(h)}) \mid \Sigma_u^{(h)}, Y &\sim \mathcal{N}(\text{vec}(\hat{A}^{(h)}), \Sigma_u^{(h)} \otimes \Omega^{(h)}).\end{aligned}$$

Inference can then be obtained by Gibbs sampling (*Inverse-Wishart posterior sampling*) at each horizon: draw $\Sigma_u^{(h,s)} \sim \mathcal{IW}(\Psi^{(h)}, \nu^{(h)})$, then draw $\text{vec}(A_0^{(h,s)}) \sim \mathcal{N}(\text{vec}(\hat{A}^{(h)}), \Sigma_u^{(h,s)} \otimes \Omega^{(h)})$. Given an identification scheme (e.g. recursive Cholesky, consistent with the baseline ordering), BLP impulse responses are computed from the posterior draws as

$$\text{BLP}(t, h, \gamma_1) = \left(A_1^{(h)}\right)^{(s)} \gamma_1,$$

i.e. the horizon- h coefficient on the impact shock scaled by the impact vector γ_1 .

A key ingredient is prior elicitation: using the LP–VAR mapping $\text{IRF}_h^{VAR} = A^h B_0^{-1}$ and $\text{IRF}_h^{LP} = A^{(h)} B_0^{-1}$,

the prior mean $A_{0,\text{prior}}^{(h)}$ can be set from iterated VAR. The BLP posterior mean is a weighted average of the OLS-LP estimate and the prior,

$$A_{\text{BLP}}^{(h)} = \left(\Omega_0^{(h)-1} + X'X \right)^{-1} \left(\Omega_0^{(h)-1} A_{0,\text{prior}}^{(h)} + X'X \hat{A}_{\text{LP}}^{(h)} \right),$$

so that tighter priors shrink BLP toward the iterated VAR benchmark, while looser priors recover the standard LP estimates.

4 Baseline Empirical Results

Figure 2 displays the responses of the patent-based innovation index and TFP to a patent-based news shock. The blue solid line is the estimated impulse response to a patent-based news shock and corresponds to the posterior median estimates. The unit of the vertical axis is the percentage deviation from the situation without a shock. The dashed red lines represent $\pm$ one standard deviation confidence bands of the patent-based news shock obtained by drawing from the posterior.

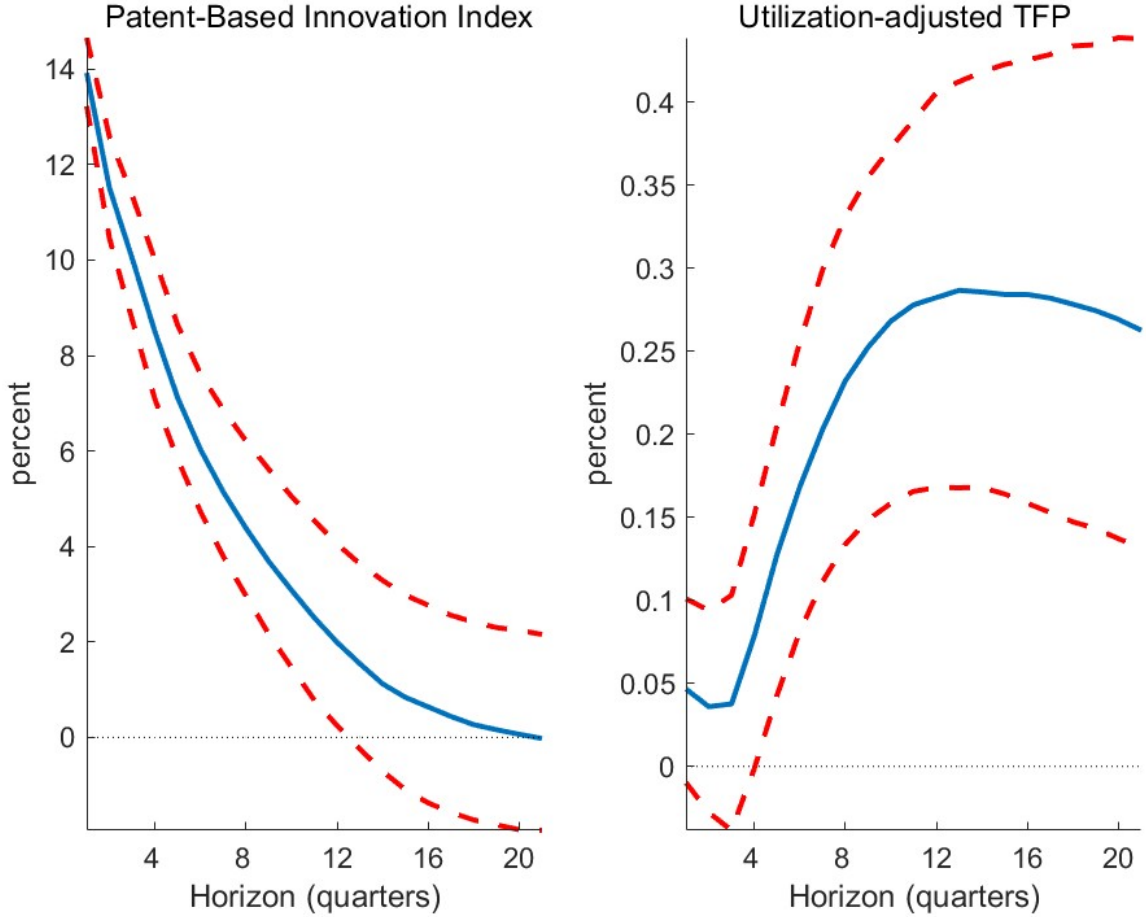


Figure 2: Responses of Patent-Based Innovation Index and TFP to a Patent-Based News Shock

After a slightly small and insignificant initial jump, TFP decreases for about a year before slowly picking up and converging to its new long-run level. This slight decrease of aggregate TFP before it starts to rise might be a reflection of a creative destruction that is also documented by KPSS at the firm level. However, it is only after about five quarters that the coverage bands of the impulse response do not rule out a zero effect. This result implies that, after an announcement of a new patent grant, a jump in the firm's stock price—a forward-looking variable—anticipates the expected future aggregate technological improvements brought by the implementation

of this patent. In fact, after a favorable patent-based news shock, TFP permanently increases by about 0.3 percent. The above result suggests that an innovation in the patent-based index carries information about aggregate productivity many periods into the future but not about its current level.

Figure 3 shows the other responses of output, consumption, investment, hours, inflation, the federal funds rate, consumer confidence, and the stock price index to a positive patent-based news shock in the model. As before, the blue solid line is the estimated impulse response to a patent-based news shock and corresponds to the posterior median estimates. The unit of the vertical axis is the percentage deviation from the situation without a shock. The dashed red lines represent $\pm$ one standard deviation confidence bands of the patent-based news shock obtained by drawing from the posterior.

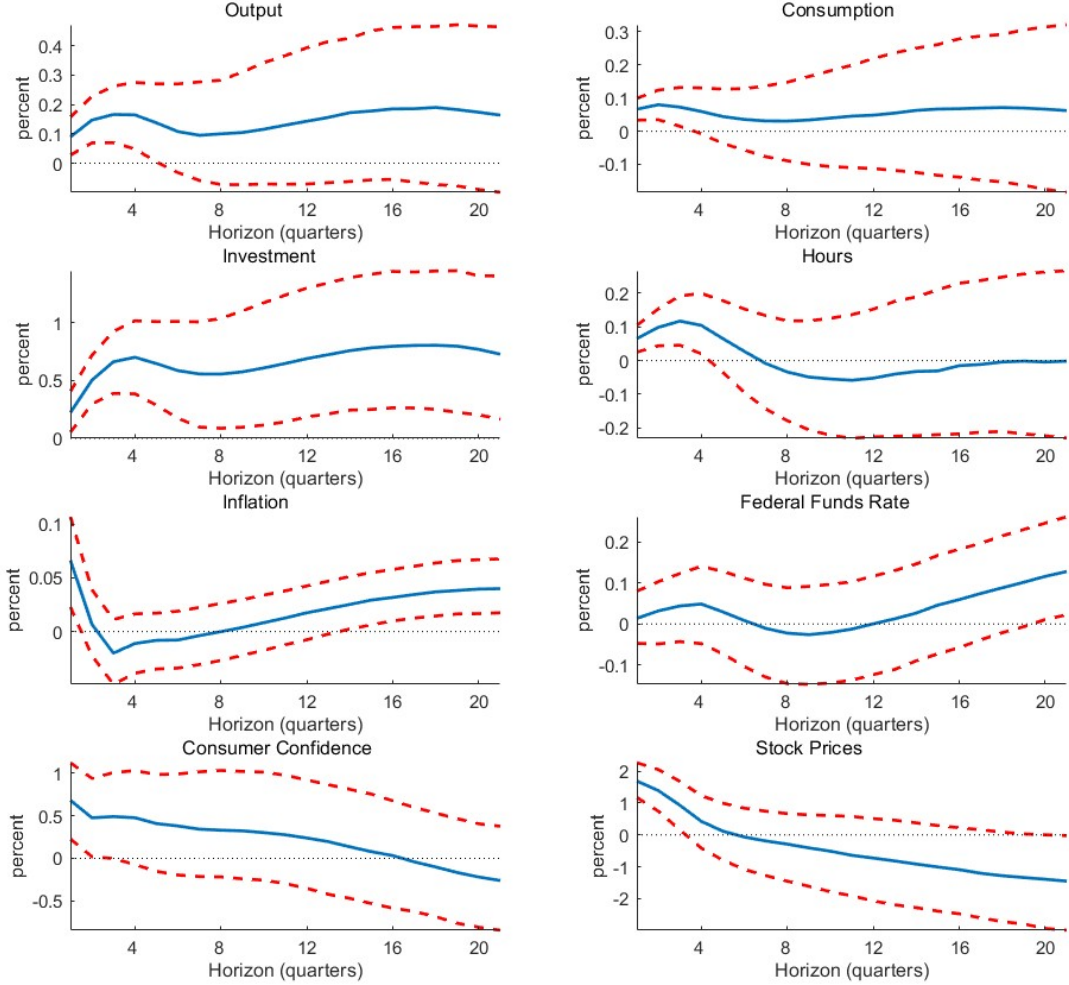


Figure 3: Responses of Patent-Based Innovation Index and TFP to a Patent-Based News Shock

A patent-based news shock induces a clear comovement among output, consumption, investment, and hours. Output increases by about 0.1 percent on impact. Interestingly, much of this increase is dictated by investment behavior. In particular, the 0.2 percent impact response of investment is higher than that of output. The effect is long-lasting as investment remains positive at a new higher level of about 0.7 percent. Consumption also reacts positively on impact, but this effect is quite muted, with an increase of about 0.1 percent. This new higher level is sustained even after five years, but the coverage bands do not rule out a zero long-run effect. The response of hours is positive on impact and through about five quarters.

Output, consumption, investment, and hours all rise on impact and display hump-shaped responses. The majority of these movements happen even before the positive effect on TFP becomes significant and TFP

starts picking up, which is clearly indicative of the identified shock carrying advance information about future productivity prospects, rather than tracking its path. The strong positive effect on impact on two forward-looking variables, stock prices and consumer confidence, also confirms the anticipation feature of the patent-based news shock. This result suggests that economic agents anticipate future technological improvements and act upon them before actual changes in TFP materialize. The impact responses of inflation and the federal funds rate are particularly interesting. While inflation rises mildly on impact, the response of the federal funds rate does not rule out a zero effect. The positive response of the federal funds rate in the short run is in line with the initial increase in inflation and with the predictions of a standard New Keynesian model.

5 Robustness Check

5.1 Patent-Based Innovation Index as an Instrument

Figure 4 presents the impulse response of the patent-based news shock on TFP. As expected, the patent-based news shock generates a permanent long-run increase in the level of TFP. There is an increase on impact, but this effect rapidly diminishes, in line with the idea of creative destruction. The shape is similar to the one presented in Figure 2 identified with the benchmark model.

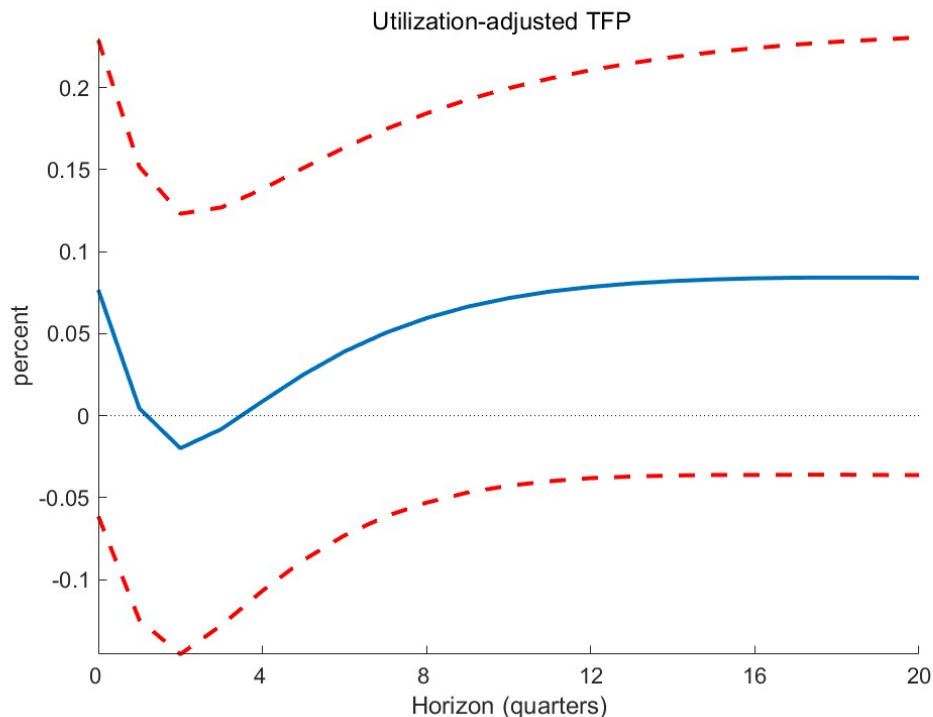


Figure 4: Response of TFP to a Patent-Based News Shock Identified with Patent-Based Innovation Index as an Instrument

Figure 5 presents the impulse responses of the other variables included in the information set. All the responses are qualitatively similar to the benchmark model shown in Figure 3. Consumption, investment, and, consequently, output jump on impact as a response to the expected future technological improvement. Hours worked react positively, converging back to zero in the long run. There is a short-run inflationary effect, which generates the expected increase in the federal funds rate. Consumer confidence and stock prices, as forward-looking variables, jump on impact, anticipating the expected future outcomes of the technological improvement. In summary, these results bring an extra confirmation of the economic effects of the patent-based news shock, in line with the theoretical idea of a neutral technological shock.

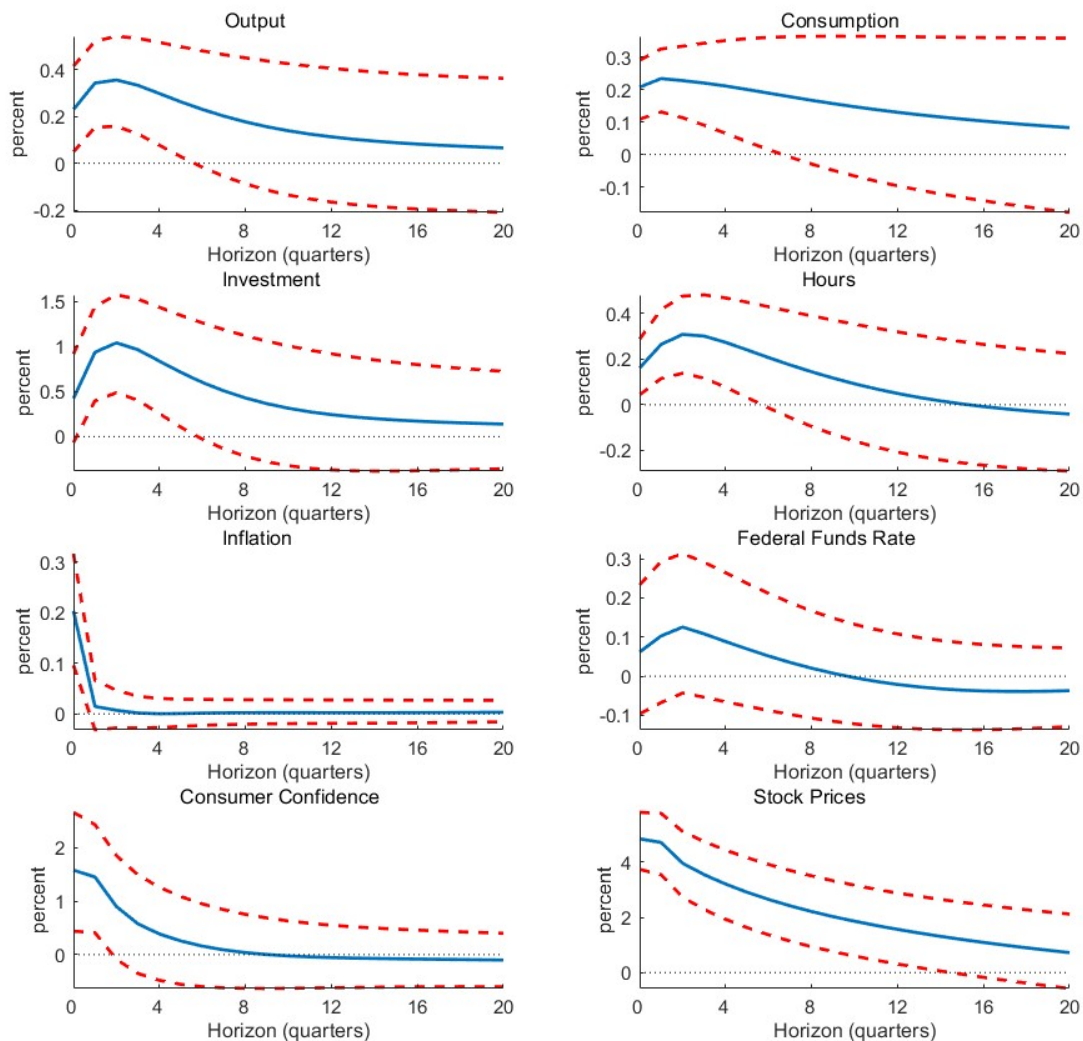


Figure 5: Responses to a Patent-Based News Shock Identified with Patent-Based Innovation Index as an Instrument

5.2 Local Projections

Figure 6 presents the impulse response of the patent-based news shock on all variables in the information set based on local projection method. The qualitative dynamics are broadly consistent with the benchmark evidence: the patent-based innovation index jumps on impact and gradually mean-reverts, while utilization-adjusted TFP increases with a delayed, persistent profile. Real activity reacts positively: investment rises strongly and output increases in a hump-shaped manner, with consumption also moving up over medium horizons. Hours worked respond positively and slowly return toward zero in the longer run. Inflation displays only modest and short-lived movements, and the federal funds rate increases accordingly. Consumer confidence and stock prices react quickly, reflecting the forward-looking nature of these variables.

A notable feature of the LP estimates, however, is that the impulse responses are not smooth across horizons: because each horizon is estimated in a separate regression, the IRFs exhibit visible horizon-to-horizon kinks and short-run oscillations (especially for inflation, confidence, and stock prices), alongside relatively wide uncertainty bands. What's more, the medium and long-run effects of the patent-based news shock, where confidence bands in local projections usually become quite wide.

5.3 Bayesian Local Projections

Figure 7 presents the corresponding Bayesian local projection impulse responses. The economic implications remain in line with the LP evidence and the benchmark narrative: the news shock raises utilization-adjusted TFP over time and generates a positive medium-run response in output, consumption, investment, and hours, while the inflation response is muted and the monetary policy rate increases in response. Forward-looking variables such as consumer confidence and stock prices react on impact, anticipating the expected future improvement in fundamentals.

Compared with standard LP, the key difference is the shape of the responses: the BLP IRFs are visibly smoother and less jagged across horizons. By shrinking the sequence of horizon-specific coefficients toward a coherent prior trajectory, Bayesian local projections reduce high-frequency horizon-to-horizon fluctuations and deliver more regular impulse response paths, while preserving the same qualitative propagation of the patent-based news shock.

6 Other useful discussions in the paper

6.1 Industry Evidence

The aggregate patent-based innovation index is based on the valuations of patents spanning many different industries. To better understand which industries serve as the wellspring of technological news, they break up the aggregate index into indexes specific to nine SIC divisions: (1) Agriculture, Forestry and Fishing; (2) Construction; (3) Finance, Insurance and Real Estate; (4) Manufacturing; (5) Mining; (6) Retail Trade; (7) Services; (8) Transportation, Communications, Electric, Gas, and Sanitary Service; and (9) Wholesale Trade. The authors disaggregate the divisions that turn out to be crucial further into industries and subindustries. For each major division (and relevant industries) they construct a patent-based innovation index following the same procedure as with the aggregate index, i.e., summing the valuations of all patents from that particular division (industry). They then repeat the same exercise as in the previous sections, replacing the aggregate index with a division (industry) index.

Figure 8 displays the responses of aggregate productivity to a positive shock to the manufacturing patent-based index (left panel) and to the electronic/electrical patent-based news shock (middle panel). In both cases, aggregate TFP does not increase on impact, takes time to become significant, and converges to a new higher value. This pattern is somewhat more pronounced when only patents in electronic/electrical industry are considered. In particular, TFP takes almost two years to become significantly different from zero, then slowly converges to a higher value, 0.4. Notice that this increase is higher than when all patents in the sample are considered. This result suggests that the patenting activity in the manufacturing sector, and in electronic/electrical industry in particular, carries important information about future movements in the aggregate TFP. The right panel of Figure 8 displays the response of the aggregate TFP to the business services patent-based news shock. TFP significantly increases on impact, but follows a similar shape as in the case of manufacturing and electronic/electrical news; it slowly decreases after the initial impact before approaching a new higher level, 0.3.

6.2 Comparing Patent-Based News Shocks with TFP-Based News Shocks

While patent-based news shocks are likely to represent a lower bound of the importance of true news shocks, the shocks identified using maximum variance decomposition of TFP are likely to represent an upper bound. It is therefore necessary to compare the recovered patent-based news shock with the most commonly used TFP-based news shock—the news shock identified using the identification proposed by Barsky and Sims (2011).

In the figure 9, The top panel displays unfiltered series of patent-based news shocks (solid blue line) and news shocks arising from the identification proposed by Barsky and Sims (2011) (dashed red line). The bottom panel represents the same two series filtered at extended business-cycle frequency (fluctuations between 6 and

40 quarters). The cyclical component is isolated using the band-pass filter. One can find that the two series are highly correlated.

What's more, illustrated in the literature, the correlation between patent-based news shocks and Barsky and Sims' news shocks is 0.61, and the correlation between our patent-based news shocks and Barsky and Sims' news shocks at extended business-cycle frequency amounts to 0.69, according to the paper, confirm that the procedure of Cascaldi-Garcia and Vukotić (2022) picks up shocks that are closely related to future improvements in technology.

7 Conclusions

Overall, the results indicate that the patent-based news shock explains a large part of longer-run variations in TFP, suggesting that it carries relevant information about future productivity movements. It also induces comovement among output, consumption, investment, and hours, but the low explanation power of movements in these variables suggests that this shock cannot be the main driver of business-cycle fluctuations. The paper represent one important step in identifying technological news shocks and understanding the macroeconomy.

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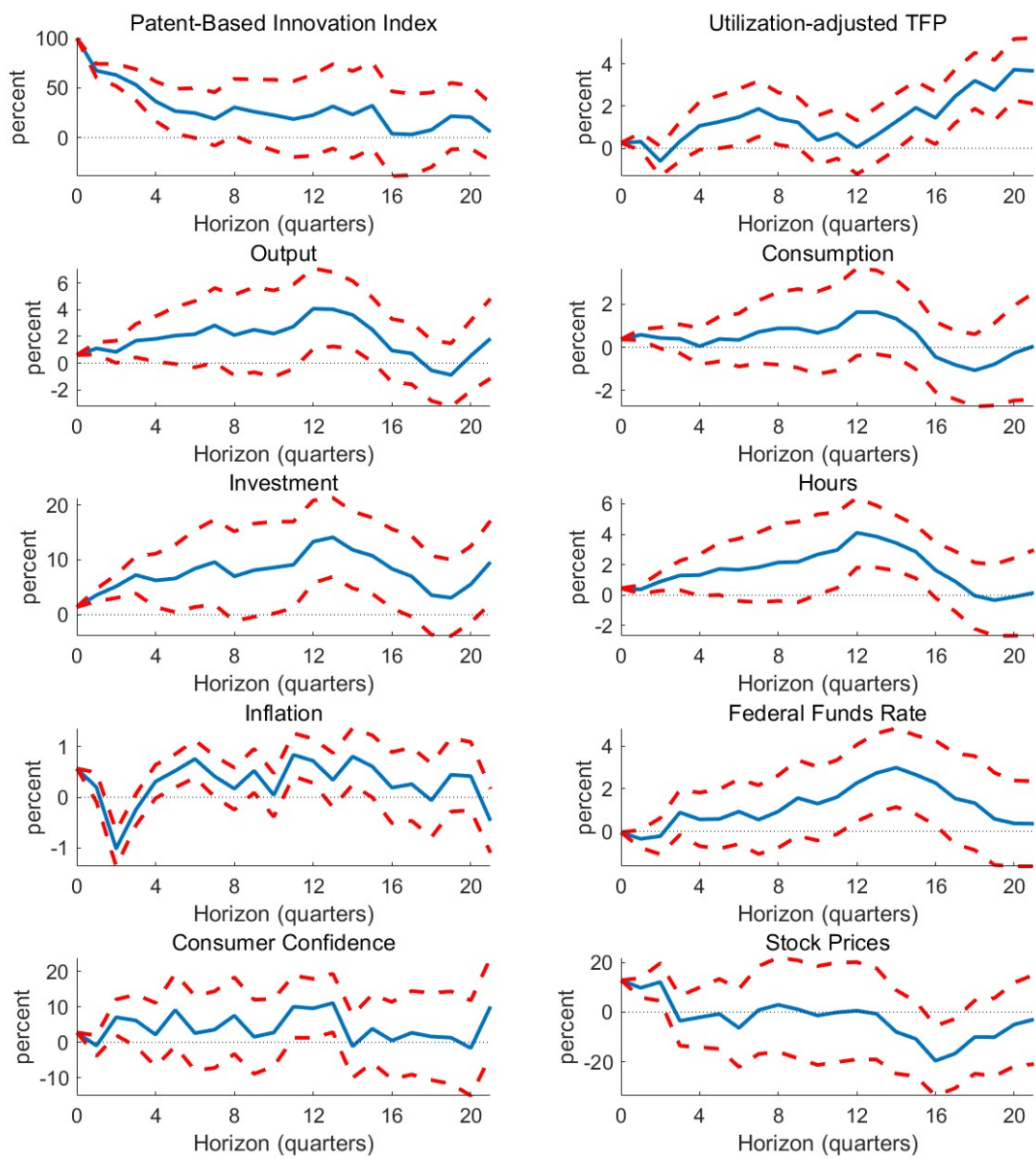


Figure 6: Local Projections of All Variables to Patent-Based News Shock

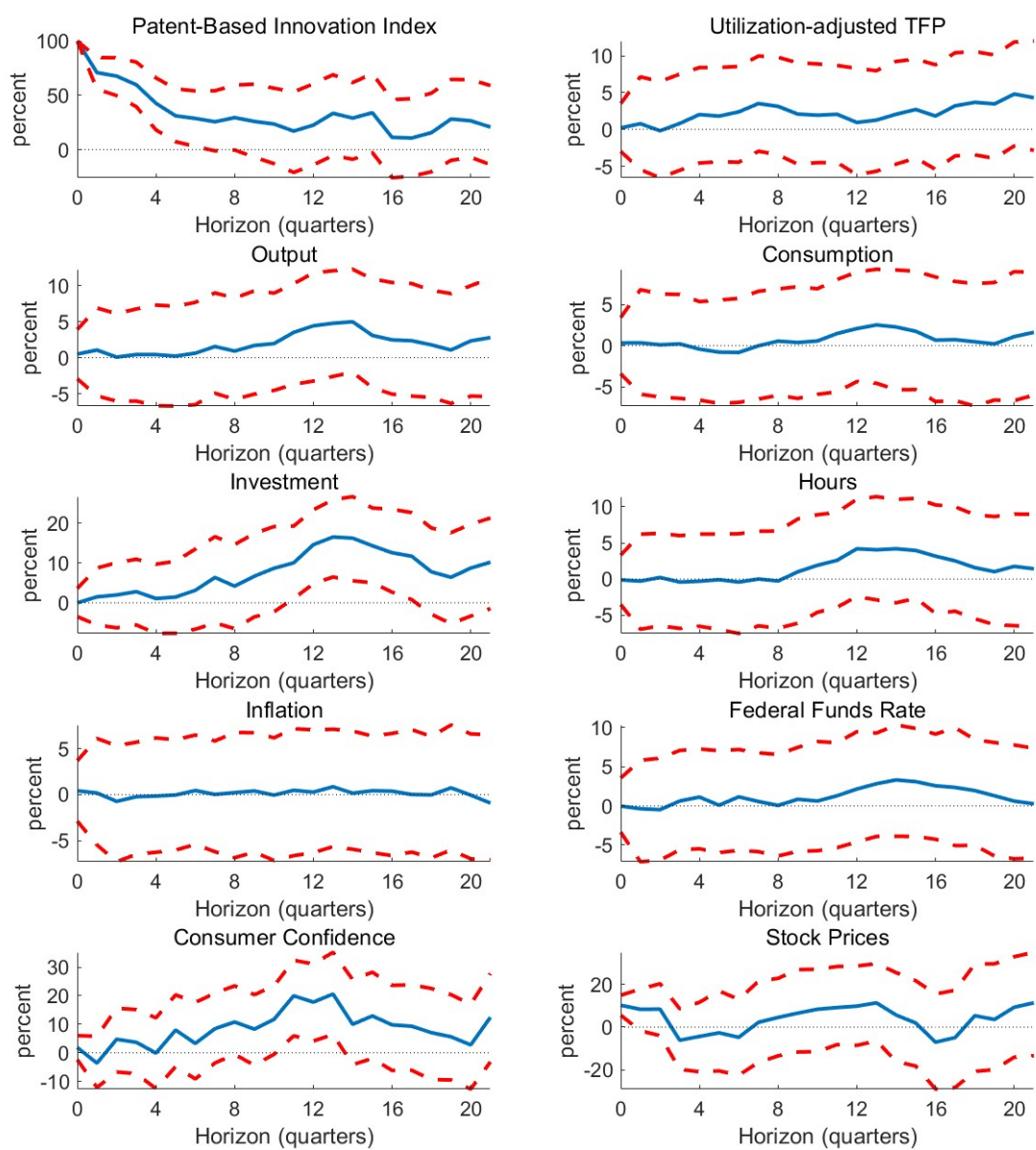


Figure 7: Bayesian Local Projections of All Variables to Patent-Based News Shock

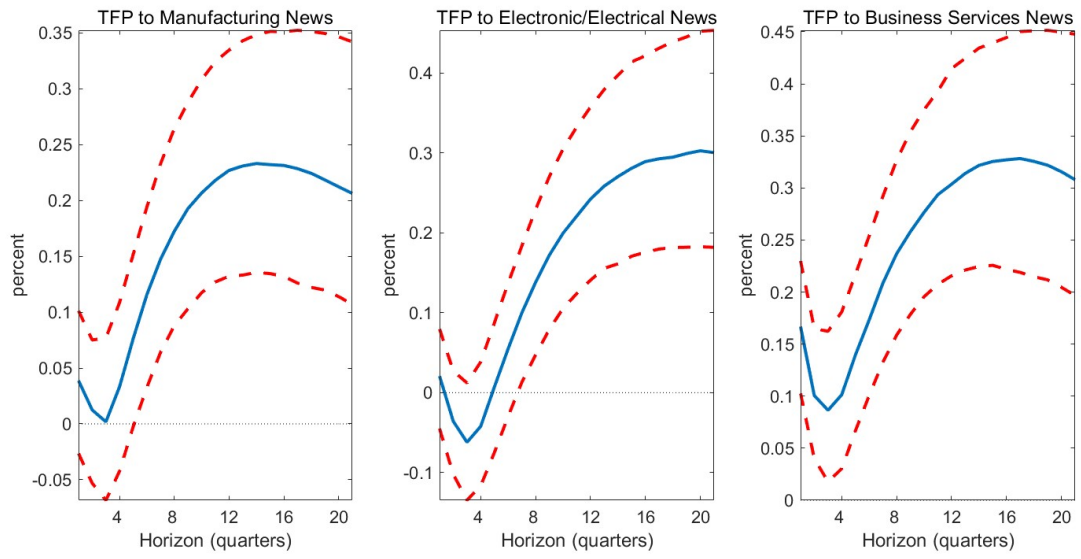


Figure 8: Effects of Disaggregated Patent-Based News Shocks on TFP

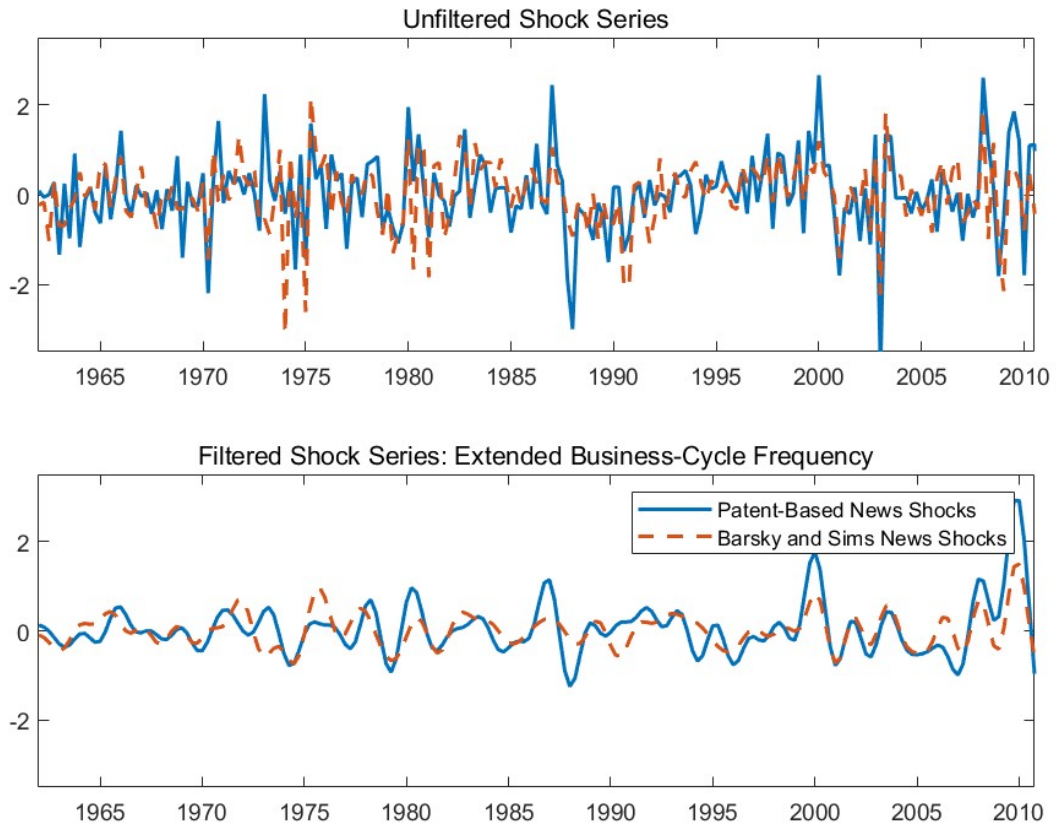


Figure 9: Patent-Based News Shocks vs TFP-Based News Shocks